

Reclaiming Our Vanishing Coastline: Why Korea Must Rethink Its Reliance Solely on Coastal Manmade Structures

Last summer, visitors to Gyeongpo Beach in Gangneung encountered an unexpected sight. The formerly expansive shoreline had significantly receded, with several sections suffering from severe erosion, and access was restricted. This phenomenon is not unique. According to the Korea Hydrographic and Oceanographic Agency, over 70 percent of the country's coastlines have experienced accelerated erosion within the past decade. Despite this reality, South Korea continues to rely predominantly on artificial structures, such as seawalls, revetments, and breakwaters, as the primary method of coastal protection. However, this structure-centric approach is increasingly unsustainable. As a researcher who has analyzed coastal sediment transport processes and coastal erosion patterns along the East Sea of Korea and contributed to a coastal erosion management project funded by the Korean Ministry of Oceans and Fisheries, I have seen how vulnerable our shorelines have become. In an era characterized by rising sea levels and intensified typhoons, it is imperative to transition toward nature-based coastal management strategies as a durable and resilient policy alternative.

Climate science has already issued a warning. The Intergovernmental Panel on Climate Change (IPCC) forecasts that global sea levels could increase by as much as one meter by the end of this century, significantly exceeding the design assumptions of many existing coastal structures. While seawalls and breakwaters are capable of diminishing wave impacts temporarily, they also interfere with littoral drift, thereby accelerating erosion in neighboring regions. An illustrative case along Korea's eastern coastline observed a reduction in beach width exceeding 20 meters within five years following the installation of a breakwater. This phenomenon is observed globally. The National Oceanic and Atmospheric Administration (NOAA) has unequivocally stated that "hard engineering alone cannot protect communities in the era of climate change."

Conversely, nature-based solutions, such as restoring wetlands, expanding salt marshes, or reconstructing dune systems, offer effective wave attenuation while maintaining ecological integrity and decreasing long-term maintenance expenses. Nations such as the Netherlands, the United States, and Australia have demonstrated significant success with these approaches, attaining both diminished erosion and enhanced flood resilience. Certainly, numerous local governments and residents advocate for artificial structures due to their provision of immediate protection. In the short term, a seawall may appear to be the most dependable barrier against rising sea levels and storm surges. However, the perceived immediacy of this protection can be deceptive. These structures frequently induce increased erosion in adjacent regions, thereby elevating the overall risk. Furthermore, maintenance and repair expenses are expected to escalate significantly as sea levels continue to rise. The stability conferred by coastal infrastructure is, in many respects, a temporal illusion.

What Korea requires at present are not bigger or higher seawalls but smarter policy decisions. The national government should integrate nature-based coastal management into its long-term climate adaptation strategy, while employing human-made coastal structures more selectively and strategically where they are genuinely necessary. Local governments must shift away from short-term, construction-focused solutions and instead invest in restoring natural systems or focus on creating environments that support climate-resilient plant species, thereby strengthening the resilience of coastal ecosystems over time. Our coastlines are more than tourist attractions: they are our frontline defense against climate-related disasters.

Across the world, it is increasingly common for nations to implement nature-based approaches to coastal management. The Netherlands' Sand Motor Project has effectively reclaimed eroded beaches, while the eastern coast of the United States has diminished storm surge impacts via marshland restoration. Korea may consider these exemplars and refer to them accordingly. Climate change remains an unavoidable reality; however, the destiny of our coastlines is within our control. The decisions made today will influence whether our coastlines disappear or are restored for future generations.